GOOGLE SUMMER OF CODE 2026 · FINAL REPORT

StablePay

AOSSIE · Djed Alliance / Stability Nexus · Amrit Kumar Yadav

1. Contributor Details

Name	Amrit Kumar Yadav
GitHub	github.com/DeveloperAmrit
Discord username	Amrit Kumar Yadav | DeveloperAmrit
Email	kittyamrit005@gmail.com
Portfolio	developeramrit.vercel.app
Pair programming peer	Atharva Naik

2. Project Details

Organization	AOSSIE (sub-organization: Djed Alliance / Stability Nexus)
Project title	StablePay
Project size	Large · 22 weeks
Primary mentor	Tanya Ruby
Mentors	Bruno, Blizet, Sarthak, Tanya, Yogesh, Nidhi, Raj, Aditya Gupta
Repositories	• StablePay (SDK & widgets) – github.com/DjedAlliance/StablePay • StablePay Merchant Dashboard – github.com/DjedAlliance/StablePay-MerchantDashboard • StablePay Landing Page – github.com/DjedAlliance/StablePay-LandingPage • Tectonic EVM Contracts – github.com/StabilityNexus/Tectonic-EVM-Contracts
Description	StablePay is a fully decentralized checkout for online payments. A merchant embeds the StablePay widget in their website; the customer signs a single transaction in their wallet, and the widget talks directly to the stablecoin protocol's smart contracts with no intermediary server, processor or custody. Customers can pay in the chain's native currency or in stablecoins, and StablePay converts automatically by minting stablecoins straight to the merchant in the same transaction. StablePay was originally built on the Djed protocol; this project improved the SDK widgets and merchant dashboard, and migrated StablePay to **Tectonic**, Stability Nexus's successor stablecoin protocol, including fixing, deploying and integrating the Tectonic contracts.

3. Summary of Work Done

The first phase of the project focused on making StablePay itself production-ready for merchants. In the StablePay SDK, I redesigned the payment widgets to a brighter theme, tightened validation and error handling across the transaction flow, and added a React + Vite example website inside the SDK that picks up every local SDK build, so widget changes can be tested end to end without republishing to npm. On the Merchant Dashboard I added paginated, sortable and progressively loaded transaction fetching (with lazy, block-grouped timestamp lookups and a non-expiring local cache), separate merchant-address settings for development and production, a working filter dialog, CSV export that respects sorting, filters and column choice, support for monitoring multiple merchant addresses, and a broader UI/UX refresh.

The second phase was the migration from Djed to Tectonic. After studying the Tectonic paper and setting the protocol up locally, I built a protocol-adapter layer in the StablePay SDK (a `ProtocolAdapter` interface with Tectonic and Djed adapters), a new `tectonic-sdk` package with the pricing math and a viem client, and a local Foundry harness with tests to develop against. Integrating against the contract surfaced real defects, so I contributed fixes upstream to Tectonic-EVM-Contracts: removing a nested reentrancy guard that made `redeem()` and forced redemptions always revert, stopping an infinitely recursive stability-fee charge, seeding both coin supplies at deployment so ratio denominators can never be zero, and handling the zero-equity state safely. After taking over the Tectonic workstream mid-term, I added a Chainlink-based inverting oracle and deployment script, deployed and verified Tectonic on the Sepolia testnet, and then switched StablePay to that public deployment, removing the local copy and the Djed integrations so StablePay works with Tectonic only.

The final phase concentrated on adoption and polish. I revamped the StablePay website with a bright design built from the StablePay and Tectonic brand palettes, new sections on payment flows, code, protocol, networks and FAQ, and Tectonic-only copy, then iterated on it with mentor feedback. I wrote full developer documentation for the website, from quick start and integration guides to widget and Tectonic SDK references, recorded video tutorials for integrating StablePay into a merchant site, and gathered scattered learning resources (Djed, Gluon, OrbOracle and others) in one place for new contributors. I also added Tectonic transaction support to the Merchant Dashboard and spent the closing weeks reviewing other contributors' PRs, getting my remaining PRs merged, and debugging StablePay end to end.

4. Week-wise Progress

Week	Description of work done	PR(s)
Community Bonding	**Task: Revise the timeline and understand Tectonic.** Revised the proposed timeline to drop the multi-Djed-variants idea in favour of studying Tectonic; got PRs from the preparation phase reviewed; reviewed open issues and PRs in the StablePay repositories.	—
Week 1	**Task: Set up StablePay repos and environment; review issues and PRs.** Set up all StablePay repositories and the local development environment, and reviewed open issues and pull requests.	—

Week	Description of work done	PR(s)
Week 2	**Task: Improve UI/UX of StablePay widgets; prevent unexpected behaviours.** Added a React + Vite example website inside `stablepay-sdk` that updates with every local SDK build; improved the design and responsiveness of the payment widgets (PricingCard, Dialog, TransactionReview); strengthened validation in the transaction flow (fixes #60, #61, #62).	StablePay #105
Week 3	**Task: Better error handling; revise widget UI/UX to a bright theme.** Added more robust error handling and user feedback during transactions and moved the widgets to a bright theme; pushed as further commits to the widget PR.	StablePay #105 (commits added)
Week 4	**Task: Performance upgrades in the merchant dashboard.** Added pagination (25/50/100/200 per page), sorting by timestamp, block number and amounts, and progressive fetching so transactions appear as they load; timestamps are fetched lazily and grouped per block, with a 1k-block cap and a "fetch more" button (fixes #31). Separated merchant-address settings for dev and prod so developers see all transactions locally. Made the persistent transaction cache non-expiring (fixes #6).	MerchantDashboard #68 MerchantDashboard #67 MerchantDashboard #71
Week 5	**Task: Transaction export with count selection and more filters; make the filter button functional.** Implemented the filter logic, copy buttons for addresses and other fields, and an improved info card; added CSV export that respects sort order, optionally applies filters, and lets the user choose columns (fixes #30, #32).	MerchantDashboard #73
Week 6	**Task: Multiple merchant address support; responsiveness, readability and UI/UX upgrades.** Merchants can now add and monitor several EVM addresses at once instead of only the connected wallet. Refreshed the dashboard into a cleaner fintech look focused on reading received payments.	MerchantDashboard #74 MerchantDashboard #75
Week 7	**Task: Read the Tectonic paper; set up Tectonic locally; note functions to import into StablePay.** Studied the Tectonic protocol, set up the contracts locally and documented the functions StablePay needs.	—
Week 8	**Task: Locally deploy Tectonic; build and test a Tectonic adapter.** Deployed Tectonic locally and built and tested the first version of the StablePay Tectonic adapter (submitted in Week 9).	—

Week	Description of work done	PR(s)
Week 9	**Task: Improve the Tectonic adapter.** Added a new `tectonic-sdk` package (pricing math + viem client); a `ProtocolAdapter` interface with TectonicAdapter and DjedAdapter that `Transaction.js` delegates to; a `protocol` field in network config; and a `tectonic-local` Foundry harness with a patched contract, mock oracle, deploy script and 21 tests.	StablePay #106
Week 10	**Task: Remove Djed from stablepay-sdk.** Removed all three Djed integrations from StablePay.	StablePay #107
Week 11	**Task: Tectonic PRs; add midterm-recommended .md files and brand folder.** Fixed Tectonic defects found during integration: a nested `nonReentrant` guard that made `redeem()` and every triggered redemption revert, a recursive `chargeStabilityFee` that ran out of gas, and an internal `ecPrice` that blocked off-chain integration. Added the repository .md files and brand folder recommended at midterm.	Tectonic-EVM-Contracts #22 StablePay #108
Week 12	**Task: Take over Tectonic work after midterm; understand progress; test and submit improvement PRs.** Seeded both coin supplies at deployment with ownerless supply so ratio denominators can never be zero, and guarded forced redemptions against an empty holder set. Handled the zero-equity state (R = L) so views report state instead of panicking, and floored the reserve ratio at the super-critical ratio; fixed the gas-diff workflow and Slither findings.	Tectonic-EVM-Contracts #24 Tectonic-EVM-Contracts #25
Week 13	**Task: Revamp the StablePay landing page; get reviews and improve Tectonic's landing page.** Replaced the dark theme with a bright design built from the StablePay and Tectonic palettes; added five sections (flows, code, protocol, networks, FAQ), the real logo mark, and Tectonic-only copy.	LandingPage #82
Week 14	**Task: Deploy Tectonic on the Sepolia testnet.** Added a Chainlink ETH/USD inverting oracle and a deployment script; deployed and verified Tectonic, EquityCoin and the oracle on Sepolia, and documented the deployment (addresses, parameters, gas notes) in Deployments.md.	Tectonic-EVM-Contracts #27
Week 15	**Task: Integrate the deployed Tectonic with StablePay; remove tectonic-local.** Switched StablePay to the Tectonic contracts deployed on Sepolia and removed the local Tectonic copy.	StablePay #111
Week 16	**Task: Implement feedback from Bruno on the StablePay website.** Iterated on the revamped website based on mentor review.	LandingPage #82

Week	Description of work done	PR(s)
Week 17	**Task: Add proper StablePay documentation to the website.** Added a docs section to the website covering introduction, quick start, how a payment works, Tectonic, integration guides (any website, Next.js), payment verification, customization, widget API, Tectonic SDK, networks, troubleshooting and glossary.	LandingPage #83
Week 18	**Task: Record video tutorials for demo StablePay integration in a merchant website.** Recorded integration video tutorials and added them to the StablePay website.	—
Week 19	**Task: Collect learning resources in one place on the website.** Gathered resources scattered across the web (Djed, Gluon, OrbOracle, etc.) into one section of the StablePay website for contributors.	LandingPage #83
Week 20	**Task: Add Tectonic transactions to the merchant dashboard.** Added an option to view Tectonic transactions; Sepolia history is read from Tenderly's public endpoint with a publicnode fallback.	MerchantDashboard #76
Week 21	**Task: Review other contributors' PRs and get remaining PRs merged.** Reviewed other contributors' pull requests and followed up on my open PRs.	—
Week 22	**Task: Debugging and troubleshooting StablePay.** Tested StablePay end to end and fixed issues found.	—